

MALWANDLA HLONGWANE

Junior Database Developer / Data-Focused ICT Graduate

Gauteng, South Africa | 065 883 1240 | MalwandlaHlongwaneICT@outlook.com | linkedin.com/in/malwandla-hlongwane | github.com/Malwandla21-ict

PROFESSIONAL SUMMARY

Final-year ICT (Applications Development) student at the University of Mpumalanga with a strong grounding in relational database design, from entity-relationship modelling and normalization through to advanced SQL, performance tuning, and emerging NoSQL/Big Data concepts. Applied this knowledge practically by designing and querying MySQL databases for two full production-style applications. Seeking a junior database developer, data analyst, or database administration role to build on this academic and applied foundation.

TECHNICAL SKILLS

- **Database Design:** Entity-Relationship (ER) modelling, normalization (up to higher normal forms), relational schema design, data modelling (conceptual, logical, physical)
- **SQL:** Data definition & manipulation, complex queries, joins, relational set operators, SQL functions, stored procedures, transaction processing
- **Database Systems:** MySQL, SQLite, Room Database (Android), MySQL Workbench
- **Advanced Topics:** Transaction management & concurrency control, database performance tuning & query optimization, distributed database systems, database security & administration
- **Emerging Data Technologies:** Big Data & Hadoop concepts, NoSQL (MongoDB, Neo4j), business intelligence & data warehousing, star schemas, SQL analytical functions
- **Applied Skills:** REST API integration with databases, CRUD operations, backend data architecture

APPLIED EXPERIENCE

Taskify: Campus Task & Equipment Marketplace

Tech: Node.js, Express, MySQL

Designed the relational database schema and query layer behind a full-stack marketplace platform.

- Designed and implemented the MySQL schema supporting users, tasks, bookings, and reviews
- Wrote CRUD queries and integrated them with a RESTful API layer
- Structured data relationships to support a full task lifecycle and equipment booking/return tracking

GitHub: github.com/Malwandla21-ict/taskify

HealthWise: AI-Based Symptom Checker App

Tech: Java, Room Database

Implemented local data persistence for a mobile health application.

- Designed and implemented a Room Database schema for storing user history and previous results

GitHub: github.com/Malwandla21-ict/HealthWise

RELEVANT COURSEWORK

- **Information Systems III (Database Systems):** ER modelling, advanced data modelling, normalization, SQL programming (beginner to advanced), relational set/join operators, database architecture (file server, client-server, web-based)
- **Information Systems III (Advanced Database Concepts):** transaction management & concurrency control, database performance tuning, distributed DBMS, database security, Big Data & NoSQL (Hadoop, MongoDB, Neo4j), business intelligence & data warehousing
- **Web Development:** MySQL database design with PHP (PDO), single- and multi-table queries, foreign keys and joins

EDUCATION

Diploma in Information and Communication Technology (Applications Development)

University of Mpumalanga, Expected Completion: 2026 | Average: 75-78%